

# Run 08 | OME

Global (including India)

IV + controls

**Target** Opportunity-motivated entrepreneurship; binary 1 vs. 0

**Sample** 1,342,884 respondents **Coverage** 89 countries; 2015–2022

**Positive cases** 100,087 (7.45%) **Case time** 38m 35s

**Train / val. / test** 913,163 / 161,145 / 268,576 (68% / 12% / 20%, stratified)

**Predictors (13):** Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.

**Controls (10):** Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.

**Protocol:** Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

## Result in brief

Highest observed ROC-AUC: Random Forest (0.9780). Highest F1: Random Forest (0.7115). For Random Forest, adding controls changed ROC-AUC by +0.0068 on the same test records.

## Model performance on held-out test records

| Model               | ROC-AUC | PR-AUC | Acc.% | Prec.% | Recall% | F1     | Time    |
|---------------------|---------|--------|-------|--------|---------|--------|---------|
| Random Forest       | 0.9780  | 0.7378 | 95.00 | 62.41  | 82.74   | 0.7115 | 2m 01s  |
| XGBoost             | 0.9779  | 0.7311 | 94.97 | 62.26  | 82.49   | 0.7096 | 23s     |
| CatBoost            | 0.9774  | 0.7240 | 94.61 | 59.52  | 86.50   | 0.7052 | 2m 49s  |
| H2O GBM             | 0.9757  | 0.7016 | 94.41 | 58.55  | 85.53   | 0.6952 | 45s     |
| Elastic Net         | 0.9738  | 0.6779 | 94.24 | 57.78  | 84.26   | 0.6855 | 11m 37s |
| Lasso               | 0.9738  | 0.6779 | 94.24 | 57.78  | 84.28   | 0.6856 | 7m 47s  |
| Logistic Regression | 0.9737  | 0.6774 | 94.25 | 57.83  | 84.31   | 0.6861 | 1m 02s  |
| Ridge               | 0.9733  | 0.6718 | 94.27 | 57.96  | 84.27   | 0.6868 | 8m 32s  |
| Decision Tree       | 0.9712  | 0.6417 | 93.87 | 55.89  | 84.40   | 0.6725 | 9s      |
| Naive Bayes         | 0.9203  | 0.4173 | 90.24 | 41.69  | 77.66   | 0.5426 | 3m 03s  |

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

## Random-forest feature importance: top 8

| Rank | Predictor                  | Raw importance | Share% |
|------|----------------------------|----------------|--------|
| 1    | Entrepreneurial intention  | 31,290.36      | 35.99  |
| 2    | Established business owner | 12,736.59      | 14.65  |
| 3    | Occupational status        | 5,027.17       | 5.78   |
| 4    | Age                        | 4,521.13       | 5.20   |
| 5    | GDP per capita (PPP)       | 4,175.16       | 4.80   |
| 6    | Population                 | 3,820.75       | 4.39   |
| 7    | GDP growth                 | 3,294.05       | 3.79   |
| 8    | Survey year                | 2,189.18       | 2.52   |

Share is the percentage of total forest impurity importance across all 40 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

**Source:** Completed Objective 1 batch, 27 September 2026; 08\_obj1\_global\_iv\_cv\_OME. Scores and importance are taken directly from saved CSV outputs.